

# Product Requirements Document

## Artist Booking SaaS Platform

For Fever Show Productions

Version 1.0 | April 16, 2026

*Confidential*

# 1. Executive Summary

This document defines the product requirements for an Artist Booking SaaS Platform designed to replace the manual, error-prone booking process currently used for repeated Fever show productions (Jury Experience, Ballet of Lights, Candlelight Concerts, etc.). The platform will centralize artist availability, automate booking suggestions, and provide real-time visibility across all stakeholders.

## 2. Problem Statement

Current Fever show booking relies on spreadsheets, email chains, and manual coordination across multiple producers, resulting in:

- Double-bookings and scheduling conflicts that go undetected until showtime
- Hours of manual effort per week matching artists to dates
- No centralized view of artist availability across shows
- Understudy assignments tracked informally with no automatic failover
- Airtable data used as source of truth but accessed inconsistently
- No audit trail for booking changes, making accountability difficult
- Scaling issues as the roster grows beyond 200+ artists across 50+ recurring shows

The platform will eliminate these inefficiencies by providing a single source of truth for all booking operations with intelligent automation.

## 3. User Personas

### 3.1 Producer / Project Manager (PM)

**Role:** Manages one or more Fever show productions and is responsible for casting artists per date.

**Goals:** Fill all dates with qualified artists quickly, minimize conflicts, ensure understudy coverage.

**Pain Points:** Manually cross-referencing availability, chasing artists via email, tracking soft-bookings in spreadsheets.

**Tech Comfort:** Moderate. Uses Airtable, Google Sheets, Slack daily.

### 3.2 Artist

**Role:** Performer booked for one or more Fever shows (musician, dancer, actor, etc.).

**Goals:** Keep availability up to date, see confirmed schedule, respond to booking requests promptly.

**Pain Points:** Forgetting to update availability, unclear booking status, last-minute cancellations without visibility.

**Tech Comfort:** Basic to moderate. Primarily mobile users.

### 3.3 Admin

**Role:** Platform administrator managing system configuration, user accounts, and data integrity.

**Goals:** Ensure platform stability, manage roles and permissions, monitor sync health, enforce booking policies.

**Pain Points:** No visibility into system health, manual user onboarding, no audit capabilities.

**Tech Comfort:** High. Comfortable with dashboards, configs, and data management.

## 4. User Stories

The following user stories are organized by persona and prioritized using MoSCoW (Must/Should/Could/Won't).

| ID     | Persona     | Story                                                                 | So That...                          | Priority |
|--------|-------------|-----------------------------------------------------------------------|-------------------------------------|----------|
| US-001 | Producer/PM | As a Producer/PM, I want to View all upcoming show dates              | I can plan assignments across shows | Must     |
| US-002 | Producer/PM | As a Producer/PM, I want to See artist availability per date          | I can assign the right artists      | Must     |
| US-003 | Producer/PM | As a Producer/PM, I want to Auto-suggest optimal artist assignments   | I save time on scheduling           | Must     |
| US-004 | Producer/PM | As a Producer/PM, I want to Override auto-suggestions manually        | I maintain creative control         | Must     |
| US-005 | Producer/PM | As a Producer/PM, I want to Soft-book an artist before confirming     | I can hold a slot while finalizing  | Must     |
| US-006 | Producer/PM | As a Producer/PM, I want to Confirm a soft-booking                    | The artist is officially assigned   | Must     |
| US-007 | Producer/PM | As a Producer/PM, I want to Assign understudies per date              | Shows have backup coverage          | Must     |
| US-008 | Producer/PM | As a Producer/PM, I want to View booking conflicts at a glance        | I can resolve double-bookings       | Must     |
| US-009 | Producer/PM | As a Producer/PM, I want to Filter shows by venue, date range, status | I find what I need quickly          | Should   |
| US-010 | Producer/PM | As a Producer/PM, I want to Export booking reports (CSV/PDF)          | I can share schedules offline       | Should   |
| US-011 | Producer/PM | As a Producer/PM, I want to Receive alerts when artists cancel        | I can find replacements fast        | Must     |
| US-012 | Producer/PM | As a Producer/PM, I want to View historical booking data              | I can analyze patterns              | Could    |
| US-013 | Artist      | As a Artist, I want to Mark my availability on a calendar             | PMs know when I can perform         | Must     |
| US-014 | Artist      | As a Artist, I want to Set recurring availability patterns            | I don't update weekly               | Should   |
| US-015 | Artist      | As a Artist, I want to View my confirmed bookings                     | I know my schedule                  | Must     |

|        |        |                                                                      |                             |        |
|--------|--------|----------------------------------------------------------------------|-----------------------------|--------|
| US-016 | Artist | As a Artist, I want to Accept or decline booking requests            | I control my commitments    | Must   |
| US-017 | Artist | As a Artist, I want to Receive notifications for new bookings        | I stay informed             | Must   |
| US-018 | Artist | As a Artist, I want to View show details (venue, time, requirements) | I can prepare               | Must   |
| US-019 | Artist | As a Artist, I want to Cancel a confirmed booking with reason        | PMs can find replacements   | Must   |
| US-020 | Artist | As a Artist, I want to See my booking history and stats              | I track my performance      | Could  |
| US-021 | Admin  | As a Admin, I want to Manage all user accounts and roles             | I control platform access   | Must   |
| US-022 | Admin  | As a Admin, I want to Configure show templates and categories        | Shows are standardized      | Must   |
| US-023 | Admin  | As a Admin, I want to View platform-wide analytics dashboard         | I monitor operations        | Must   |
| US-024 | Admin  | As a Admin, I want to Configure Airtable sync settings               | Data stays current          | Must   |
| US-025 | Admin  | As a Admin, I want to Manage notification preferences globally       | Communication is controlled | Should |
| US-026 | Admin  | As a Admin, I want to Audit trail of all booking changes             | I maintain accountability   | Must   |
| US-027 | Admin  | As a Admin, I want to Bulk import/export artist data                 | I manage large rosters      | Should |
| US-028 | Admin  | As a Admin, I want to Set booking rules and constraints              | Scheduling follows policies | Should |
| US-029 | Admin  | As a Admin, I want to Monitor system health and sync status          | I ensure uptime             | Should |
| US-030 | Admin  | As a Admin, I want to Manage venue and show metadata                 | Data is accurate            | Must   |

## 5. Feature List (MoSCoW Prioritization)

| Feature                         | Description                                                                             | Priority   |
|---------------------------------|-----------------------------------------------------------------------------------------|------------|
| Authentication & RBAC           | Role-based access (Admin, Producer/PM, Artist) with Supabase Auth + RLS                 | Must       |
| Shows & Dates CRUD              | Create, read, update, delete shows and their date instances                             | Must       |
| Airtable Sync Engine            | Edge function polling Airtable with configurable interval + local cache fallback        | Must       |
| Artist Availability Calendar    | Calendar UI for artists to mark available/unavailable per date                          | Must       |
| Auto-Suggest Booking Engine     | Algorithm to suggest optimal artist assignments based on availability, priority, skills | Must       |
| Manual Override                 | PM can override auto-suggestions with manual artist selection                           | Must       |
| Soft-Book / Confirm Workflow    | Two-step booking: soft-book (hold) then confirm (lock)                                  | Must       |
| Understudy Management           | Assign and auto-promote backup artists per date                                         | Must       |
| Conflict Detection              | Real-time detection of double-bookings and scheduling conflicts                         | Must       |
| Role-Specific Dashboard         | Admin: platform-wide; PM: their shows; Artist: their bookings                           | Must       |
| In-App Notifications            | Alerts for new dates, booking confirmations, cancellations, schedule changes            | Should     |
| Booking History & Audit Trail   | Full history of all booking state changes with timestamps                               | Should     |
| Recurring Availability Patterns | Artists set weekly/monthly patterns instead of individual dates                         | Should     |
| CSV/PDF Export                  | Export booking reports and schedules                                                    | Should     |
| Venue Management                | CRUD for venues with capacity, location, technical specs                                | Should     |
| Search & Filtering              | Advanced search across shows, artists, dates with multiple filters                      | Should     |
| Artist Skills & Tags            | Tag artists with skills/instruments for smarter suggestions                             | Could      |
| Analytics & Reporting           | Time-to-book, fill rate, understudy usage, artist utilization                           | Could      |
| Bulk Operations                 | Bulk availability updates, bulk booking confirmations                                   | Could      |
| Mobile-Responsive UI            | Full functionality on mobile devices                                                    | Won't (v1) |



## 6. Data Model

The following entities form the core data model, implemented in Supabase (PostgreSQL) with Row-Level Security.

### 6.1 Shows

- `id` (UUID, PK) — Unique identifier
- `title` (text) — Show name (e.g., Jury Experience)
- `description` (text) — Show description
- `venue_id` (UUID, FK) — Reference to venue
- `category` (text) — Show type (concert, theater, dance, etc.)
- `requirements` (jsonb) — Artist requirements (skills, instruments, etc.)
- `status` (enum) — active, archived, draft
- `created_by` (UUID, FK) — User who created the show

### 6.2 Show Dates

- `id` (UUID, PK) — Unique identifier
- `show_id` (UUID, FK) — Reference to parent show
- `date` (date) — Performance date
- `start_time` (time) — Start time
- `end_time` (time) — End time
- `slots_required` (int) — Number of artists needed
- `status` (enum) — open, partially\_filled, filled, cancelled
- `airtable_record_id` (text) — Synced Airtable record ID

### 6.3 Artists (Profiles)

- `id` (UUID, PK) — Matches `auth.users` ID
- `display_name` (text) — Stage/display name
- `email` (text) — Contact email
- `phone` (text) — Contact phone
- `skills` (text[]) — Array of skills/instruments
- `priority_score` (int) — Booking priority (higher = preferred)
- `status` (enum) — active, inactive, on\_leave

### 6.4 Availability

- `id` (UUID, PK) — Unique identifier
- `artist_id` (UUID, FK) — Reference to artist
- `date` (date) — Available date
- `status` (enum) — available, unavailable, tentative
- `recurrence_rule` (text) — iCal RRULE for recurring patterns

## 6.5 Bookings

- `id` (UUID, PK) — Unique identifier
- `show_date_id` (UUID, FK) — Reference to show date
- `artist_id` (UUID, FK) — Reference to artist
- `status` (enum) — suggested, soft\_booked, confirmed, cancelled
- `is_understudy` (boolean) — Whether this is an understudy assignment
- `booked_by` (UUID, FK) — PM who created the booking
- `confirmed_at` (timestampz) — When booking was confirmed
- `cancelled_at` (timestampz) — When booking was cancelled
- `cancellation_reason` (text) — Reason for cancellation

## 6.6 Booking Audit Log

- `id` (UUID, PK) — Unique identifier
- `booking_id` (UUID, FK) — Reference to booking
- `action` (text) — created, soft\_booked, confirmed, cancelled, understudy\_promoted
- `performed_by` (UUID, FK) — User who performed the action
- `details` (jsonb) — Additional context

## 6.7 Airtable Sync Log

- `id` (UUID, PK) — Unique identifier
- `sync_type` (text) — poll, webhook, manual
- `status` (text) — success, failure, partial
- `records_processed` (int) — Number of records synced
- `error_details` (text) — Error message if failed
- `synced_at` (timestampz) — When sync occurred

## 7. Airtable Sync Design

### 7.1 Architecture

The platform uses a hybrid polling + webhook approach to sync show dates from Airtable into the Supabase database.

#### Primary: Polling via Edge Function

- A Supabase Edge Function runs on a configurable interval (default: 5 minutes)
- Queries Airtable API for new/modified records since last sync timestamp
- Upserts records into show\_dates table using airtable\_record\_id as the match key
- Logs each sync attempt to airtable\_sync\_log for monitoring

#### Secondary: Webhook Listener

- An Edge Function endpoint receives Airtable webhook payloads
- Processes changes in real-time when webhooks are configured
- Falls back to polling if webhook delivery fails

### 7.2 Caching & Offline Resilience

- Last-known-good data is always stored in Supabase tables
- If Airtable API is unreachable, the platform continues operating from cached data
- Sync status is displayed in the Admin dashboard with health indicators
- Cache TTL is configurable (default: 24 hours before data is considered stale)

### 7.3 Conflict Resolution

- Airtable is treated as the source of truth for show dates
- Local modifications to synced fields are overwritten on next sync
- Booking data (assignments, availability) lives exclusively in Supabase

## 8. Booking Workflow

### 8.1 Hybrid Auto-Suggest + Manual Override Flow

The booking engine follows a multi-step workflow combining automation with human oversight:

**Step 1 — Date Release:** New show dates appear (via Airtable sync or manual creation). Status: OPEN.

**Step 2 — Availability Check:** System queries artist availability for the date. Artists with status = 'available' and matching skills are eligible.

**Step 3 — Auto-Suggest:** Algorithm ranks eligible artists by: (a) priority\_score, (b) recent booking frequency (load balancing), (c) skill match score, (d) historical reliability. Top N suggestions are presented to the PM.

**Step 4 — PM Review:** PM reviews suggestions and can: Accept suggestion (status → SOFT\_BOOKED), Override with different artist, Skip and handle manually later.

**Step 5 — Soft-Book:** Artist receives notification. Can accept or decline within configurable window (default: 48 hours). If no response, PM is alerted.

**Step 6 — Confirm:** PM confirms the booking. Status → CONFIRMED. Artist receives confirmation notification. Audit log entry created.

**Step 7 — Understudy:** PM assigns understudy. If primary cancels, understudy is auto-promoted to CONFIRMED and notified.

### 8.2 Cancellation Flow

- Artist or PM initiates cancellation with required reason
- If understudy exists: auto-promote understudy to primary, notify all parties
- If no understudy: alert PM, date status reverts to OPEN for that slot
- All cancellations logged to audit trail

## 9. Wireframe Descriptions

### 9.1 PM Dashboard

- Top bar: Show selector dropdown, date range filter, notification bell
- Main area: Calendar/timeline view showing all dates for selected show
- Each date card shows: date, time, venue, slots filled/total, status badge
- Click date → opens booking panel with suggested artists and current assignments
- Right sidebar: Quick stats (fill rate, pending soft-books, upcoming dates)

### 9.2 Booking Panel (Date Detail)

- Header: Show name, date, venue, time
- Section 1: Current Assignments — list of confirmed/soft-booked artists with status badges
- Section 2: Auto-Suggestions — ranked list of available artists with match scores
- Section 3: Manual Search — search/filter all artists with availability overlay
- Action buttons: Soft-Book, Confirm, Replace, Assign Understudy

### 9.3 Artist Calendar View

- Monthly calendar grid with color-coded days
- Green = available, Red = unavailable, Blue = booked, Yellow = soft-booked
- Click day → toggle availability or view booking details
- Sidebar: Upcoming bookings list with show details
- Top: Quick-set recurring availability (e.g., available every weekday)

### 9.4 Admin Control Panel

- System Health: Airtable sync status, last sync time, error count
- User Management: User list with role badges, invite/deactivate actions
- Show Management: Show list with date counts, fill rates
- Audit Log: Searchable/filterable log of all booking changes
- Configuration: Sync interval, cache TTL, booking rules, notification settings

## 10. Success Metrics

| Metric                | Current State              | Target (v1)          | Measurement                                        |
|-----------------------|----------------------------|----------------------|----------------------------------------------------|
| Time-to-Book (avg)    | 2-4 hours per show date    | < 15 minutes         | Timestamp diff: date created → all slots confirmed |
| Fill Rate             | ~85% (manual tracking)     | 95%+                 | Confirmed slots / required slots across all dates  |
| Understudy Coverage   | ~30% of dates              | 90%+ of dates        | Dates with understudy / total dates                |
| Double-Booking Rate   | ~5% discovered at showtime | 0%                   | Conflict detection alerts / total bookings         |
| Artist Response Time  | 48-72 hours (email)        | < 24 hours           | Soft-book → accept/decline timestamp               |
| Airtable Sync Lag     | Manual (hours/days)        | < 5 minutes          | Last sync timestamp freshness                      |
| PM Satisfaction (NPS) | Not measured               | > 40                 | Quarterly survey                                   |
| Artist Adoption Rate  | N/A (new platform)         | 80%+ within 3 months | Active artists / total invited                     |

## 11. Technical Architecture

### 11.1 Stack

- Frontend: React 18 + TypeScript + Vite + Tailwind CSS
- Backend: Supabase (PostgreSQL, Auth, Edge Functions, Storage)
- External Integration: Airtable API (read-only sync)
- Hosting: Lovable Cloud / Supabase

### 11.2 Security

- Row-Level Security (RLS) on all tables
- Role-based access control via user\_roles table (admin, producer, artist)
- JWT validation in Edge Functions
- All Airtable API keys stored as Supabase secrets

### 11.3 Folder Structure

src/

- config/ — app.config.ts (feature flags, API endpoints, sync intervals, cache TTLs)
- features/ — auth/, shows/, artists/, bookings/, availability/, understudy/, sync/, dashboard/, notifications/
- hooks/ — Shared React hooks
- services/ — API service layers
- types/ — Shared TypeScript types
- components/ — Shared UI components

## 11.4 Central Configuration

All configurable parameters are defined in src/config/app.config.ts with full TypeScript typing and JSDoc comments. This includes:

- Feature flags (enable/disable modules)
- Airtable sync interval (default: 5 min)
- Cache TTL (default: 24 hours)
- Soft-book response window (default: 48 hours)
- Auto-suggest algorithm weights
- Notification preferences
- Role definitions and permissions

## 12. Risks & Mitigations

| Risk                                      | Impact | Mitigation                                                  | Owner        |
|-------------------------------------------|--------|-------------------------------------------------------------|--------------|
| Airtable API rate limits                  | High   | Implement caching, batch requests, respect rate headers     | Engineering  |
| Artist adoption resistance                | High   | Intuitive UX, mobile-first calendar, gradual rollout        | Product + PM |
| Data conflicts between Airtable and local | Medium | Clear source-of-truth rules, conflict logging, admin alerts | Engineering  |
| Edge function cold starts                 | Low    | Keep functions warm, optimize bundle size                   | Engineering  |
| Scale beyond 200+ artists                 | Medium | Pagination, efficient queries, indexing strategy            | Engineering  |

## 13. Release Plan

### Phase 1 — Foundation (Weeks 1-4)

- Auth & Roles, Show/Date CRUD, Artist Profiles
- Basic availability calendar
- Airtable sync (polling only)

### Phase 2 — Core Booking (Weeks 5-8)

- Auto-suggest engine
- Soft-book / confirm workflow
- Understudy management
- Conflict detection

### Phase 3 — Polish & Scale (Weeks 9-12)

- Notifications system
- Role-specific dashboards
- Analytics & reporting
- Webhook-based Airtable sync
- Export capabilities

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